

Human–robot negotiation in constrained shared spaces under uncertain human intentions

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Background

In robotic navigation scenarios, reaching the target without collision does not necessarily mean that the robot behaves appropriately around humans. In constrained shared spaces such as narrow corridors and doorways, the robot may need to decide whether to proceed, slow down, wait, or yield when encountering pedestrians. Such decisions depend not only on instant pedestrian positions and velocities, but also on their intentions and reactions. For example, one pedestrian may slow down and allow the robot to pass and may change his mind because he thinks that the robot is moving at a slow speed. Since these intentions are not directly observable and may change in response to the robot's actions, socially acceptable navigation becomes an interactive decision-making problem rather than simple collision avoidance.

Conventional Algorithms

Traditional robot navigation is commonly divided into a global planner, such as A*, which determines a macro route through the static environment, and a local planner based on Social Force Model and velocity-obstacle-based methods (ORCA) that handles close pedestrians.

However, such approaches become difficult when navigation requires interaction with pedestrians whose intentions are uncertain. A fixed avoidance rule may cause unnecessary waiting when a pedestrian is willing to yield, while an aggressive rule may create conflicts with a pedestrian who expects to pass first. Predicting pedestrian motion independently of the robot may be insufficient because the pedestrian will also react to what the robot does. Handling these situations conventionally therefore requires explicit assumptions.

We believe that reinforcement learning provides an alternative in which the robot can learn interaction strategies through experience. By observing how pedestrians react to its previous actions, the robot can learn when to proceed, wait, or yield without requiring their intentions to be directly specified. The objective of this project is to investigate whether reinforcement learning can develop effective implicit negotiation behavior under uncertain pedestrian intentions.

Problem Statement:

A discrete grid world with the reachable grids shaping 'T' (within $N \times M$ cells) containing static obstacles, one pedestrian, a single robot, and a goal. The robot and the pedestrian are moving towards each other along the horizontal corridor, and they are both aiming for a door located at the joint of the T-shaped world to enter. Robot's task is to reach the goal at the bottom of the vertical corridor, while resolving the doorway conflict without collision or exceeding a predefined interaction timeout.

The robot can observe the ($N \times M$) map and the current position (2D vector containing the current position) of the pedestrian. But the pedestrian's underlying intention is not directly observable. A one-hot vector representing different hidden behavioral patterns, such as yielding or continuing to move, is unknown to the robot. The robot therefore needs to infer their behavior from recent interactions and learn when to proceed, wait, or yield.

RL Cast:

1\ state space: Four $N \times M$ arrays showing the robot position, static obstacles, pedestrians, and the goal position.

2\ action space: 13 actions. In addition to the five basic actions (move North, East, South, West, or stay in place), the robot can perform eight two-step movements: four moving two cells in the same direction and four combining two perpendicular directions. The extended action space allows the robot to express different levels of movement commitment.

3\ reward structure: A positive reward for reaching the goal (say 10), a large penalty for collision (-10), and a small step penalty to discourage unnecessary movement and excessive waiting.

4\ NN structure: A CNN extracts spatial features from the grid observation, followed by an LSTM to capture recent human-robot interactions. Fully connected layers then produce the policy and value estimates.